

Wireless Multicam Studio

Turn your laptop into a multi-camera studio and a phone into a wireless camera, in about 10 minutes.

What this is	An open-source Windows tool that links phones and webcams into one synchronized multi-camera recording rig over your local WiFi.
On the phone	No app to install. Phones connect through their normal web browser (Safari on iPhone, Chrome on Android).
Your data	Everything stays on your network. No internet connection, no account, and nothing is uploaded to the cloud.
Time needed	About 10 minutes for first setup. Later sessions take under a minute.

What you need

- ✓ **A Windows 10/11 laptop or desktop.**
This is the machine that runs the studio.
- ✓ **One or more phones.**
iPhone with Safari, or Android with Chrome. No app install needed.
- ✓ **All devices on the SAME WiFi / network.**
No internet connection is required.
- ✓ **(Optional) a webcam on the laptop.**
Handy if you want a second angle without using a phone.

Step-by-step

1 Install the studio — on the laptop

Download **WirelessMulticamStudio-Setup.exe** (from the website's **Download** button or the GitHub Releases page) and run it.

Windows SmartScreen will warn that it's unsigned — click **More info**, then **Run anyway**. It's safe; it's just not code-signed yet.

It installs and adds Start Menu and desktop shortcuts. Launch **Wireless Multicam Studio**. It runs quietly as a system-tray icon (bottom-right of the taskbar) — there is no app window.

2 One-time HTTPS setup — on the laptop

The studio needs a local security certificate so phones are allowed to use their cameras.

If you didn't run it during install: right-click the tray icon and choose **First-time HTTPS setup**. Approve the Windows admin prompt (UAC).

You only do this once per laptop.

After the certificate is set up, you won't be prompted again on this machine.

3 Open the Operator Dashboard — on the laptop

Right-click the tray icon and choose **Open Operator Dashboard**. It opens at **https://studio.localhost:8444/**. This is your control room — every camera shows up here.

Optional — add the laptop's own webcam

Right-click the tray icon and choose **Open Device Page**, then join as a camera from the laptop to use it as an extra angle.

4 Connect a phone as a camera

4a Get the device link onto the phone. The phone must be on the same WiFi.

Easiest: right-click the tray icon → **Show Device QR (scan to connect)** and scan the QR code with the phone's camera.

Or: right-click the tray icon → **Show URLs** and type the **Devices** address into the phone's browser.

4b Install the certificate on the phone — one time per phone. The device page has a guided **Download certificate** button with step-by-step instructions and a QR code. In short:

iPhone (iOS): open/download the certificate and install the profile, then go to **Settings** →

General → **About** → **Certificate Trust Settings** and turn ON full trust for the certificate. Both steps are required, or the camera stays blocked.

Android: download the certificate, then **Settings** → **Security** → **Install a certificate** → **CA certificate**.

4c Join. Reload the device page, type a name (e.g. "Stage Cam"), pick a source (**Camera** , **Screen share** , or **Audio-only**), and tap **Join** .

For a phone camera, turn the phone to landscape.

5 Record a quick test — on the Operator Dashboard

Under **Cameras** , add a camera and assign your joined phone to it.

Click **Start Preview (all)** — the live feed appears in the grid. Frame your shot.

(Optional) click **Pre-flight** to confirm every camera is live, has audio, and the disk is writable.

Click the red ● **Record** button — every angle records at full quality.

While recording, click a feed tile (or press number keys **1–9**) to mark which camera is the **program "take"**. The switch log records every cut.

Click **Stop Recording** when you're done.

6 Get your recordings

Click **Recordings** in the header to browse and download the per-camera files and the switch log, preview the program edit in the browser, or **Export** the whole thing as one finished MP4.

Tips & troubleshooting

Nothing leaves your network

Everything stays on your WiFi — no internet, no account, nothing uploaded.

Camera blocked on the phone?

The certificate isn't trusted yet — re-check Step 4b. On iPhone the camera is silently blocked until the cert is fully trusted.

Changed WiFi networks?

Just relaunch the studio; the phone stays trusted — no need to reinstall the certificate.

Security: LAN-only, no password

Anyone on the same WiFi who opens the operator address can control it. Test on a network you trust.